

Grayness

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This Typst package provides basic image editing functions like grayscaling, inverting, cropping and applying color transforms. Moreover, this package supports image-formats not natively available in Typst. The following formats can be used:

- | | | | |
|------------|-----------|--------|---------|
| • BMP | • HDR | • PNG* | • TIFF |
| • DDS | • ICO | • PNM | • WebP* |
| • Farbfeld | • JPEG* | • QOI | • SVG* |
| • GIF* | • OpenEXR | • TGA | |

* Natively supported by Typst

All examples in this manual use one of the following images as their base:



WebP: Pianist [Arturo Nieto Dorantes](#),
[CC BY-SA 4.0](#) by Laëtitia Boudaud



SVG: A traced [Lamborghini Gallardo](#),
[CC BY-NC-SA 2.5](#) by Michael Grosberg

Available functions

- `image-grayscale()`
- `image-show()`
- `image-crop()`
- `image-flip-horizontal()`
- `image-flip-vertical()`
- `image-blur()`
- `image-transparency()`
- `image-brighten()`
- `image-darken()`
- `image-invert()`
- `image-matrix()`
- `image-huerotate()`
- `image-mask()`
- `image-infos()`

Variables

- `plg`

image-grayscale

Create a grayscale-image representation of the provided imagedata (Raster or SVG)

Example:

```
#import "@preview/grayness:0.8.0": *  
#let arturo = path("Arturo_Nieto-Dorantes.webp")  
#image-grayscale(arturo)
```



Parameters

```
image-grayscale(  
  imagedata: bytes path ,  
  ..args  
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to the file

..args

extra arguments to pass to the Typst image function e.g. width, height, format, etc...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

Example:

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-grayscale(gallardo, format:"svg",  
width:4cm, alt:"Lamborghini Gallardo")
```



image-show

Displays an image from raw bytes or a path to a file. This enables the usage of imageformats not natively supported by Typst

Example:

```
#import "@preview/grayness:0.8.0": *  
#let arturo = path("Arturo_Nieto-Dorantes.webp")  
#image-show(arturo)
```



Parameters

```
image-show(  
  imagedata: bytes ,  
  ..args  
) -> content
```

imagedata bytes

Raw imagedata, e.g. provided by the `read()` function

..args

extra arguments to pass to the Typst image function e.g. width, height, format, etc...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

Example:

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-show(gallardo, format:"svg",  
width:2cm, alt:"Lamborghini Gallardo")
```



image-crop

Crop the given imagedata to the specified width and height

Example:

```
#import "@preview/grayness:0.8.0": *
#let arturo = read("Arturo_Nieto-
Dorantes.webp", encoding: none)
#image-crop(
  arturo,
  crop-width: 100,
  crop-height: 120,
  crop-start-x: 190,
  crop-start-y: 95
)
```



Parameters

```
image-crop(
  imagedata: bytes path,
  crop-width: int,
  crop-height: int,
  crop-start-x: int,
  crop-start-y: int,
  ..args: arguments
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

crop-width int

Horizontal size (in pixels or “user space” in case of SVG) of the crop window. Must be ≥ 0 .

Default: 10

crop-height int

Vertical size (in pixels or “user space” in case of SVG) of the crop window. Depending on the settings of the SVG, it might preserve its aspect ratio even if crop-width and crop-height have a different ratio. Must be ≥ 0 .

Default: 10

crop-start-x int

Left starting coordinate (in pixels or “user space” in case of SVG) of the crop window. Depending on the settings of the SVG, it might preserve its aspect ratio even if crop-width and crop-height have a different ratio. Must be ≥ 0 .

Default: 0

crop-start-y `int`

Top starting coordinate (in pixels or “user space” in case of SVG) of the crop window. Must be ≥ 0 .

Default: `0`

..args `arguments`

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format: "svg"` as argument if you use a SVG-image as your input.

Example:

```
#import "@preview/grayness:0.8.0": *
#let gallardo = path("gallardo.svg")
#image-crop(
  gallardo,
  crop-height: 200,
  crop-width: 100,
  crop-start-x: 10,
  crop-start-y: 250,
  format: "svg"
)
```

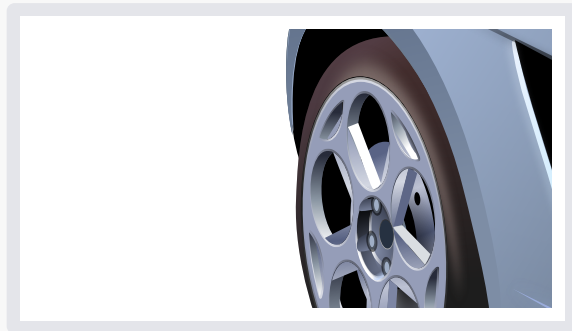


image-flip-horizontal

Flip the provided imagedata horizontally *Example:*

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-flip-horizontal(gallardo,  
  format:"svg")
```



Parameters

```
image-flip-horizontal(  
  imagedata: bytes path ,  
  ..args  
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

..args

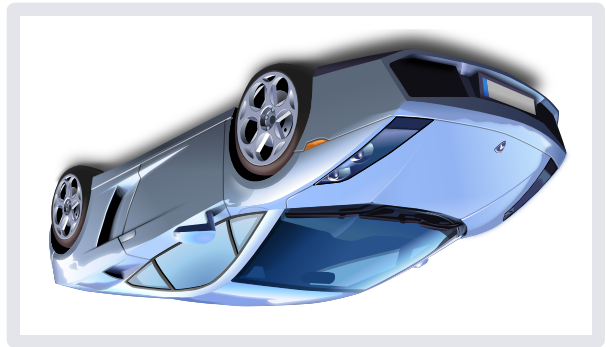
Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

image-flip-vertical

Flip the provided imagedata vertically *Example:*

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-flip-vertical(gallardo,  
format:"svg")
```



Parameters

```
image-flip-vertical(  
  imagebytes,  
  ..args  
) -> content
```

image-blur

Performs a Gaussian blur on the imagedata.

Warning: This operation is slow, especially for large sigmas.

```
#import "@preview/grayness:0.8.0": *  
#let arturo = path("Arturo_Nieto-Dorantes.webp")  
#image-blur(arturo, sigma:3.14)
```



Parameters

```
image-blur(  
  imagedata: bytes path,  
  sigma: int float,  
  ..args  
) -> content
```

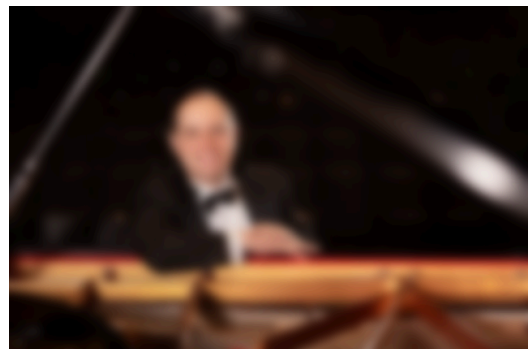
imagedata bytes or path

Raw imagedata in any of the supported formats, e.g. provided by the `read()` function or a path to a file

sigma int or float

A measure of how much to blur by (standard deviation)

```
#import "@preview/grayness:0.8.0": *  
#let arturo = read("Arturo_Nieto-Dorantes.webp", encoding: none)  
#image-blur(arturo, sigma:6.28)
```



Default: 5

..args

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-blur(gallardo, sigma: 40, format:  
"svg")
```



image-transparency

Adds transparency to the provided image data

Example:

```
#import "@preview/grayness:0.8.0": *  
//Place rectangle in background to  
demonstrate transparency  
#place(top+center, dx:-3cm)  
[#rect(fill:blue, width:4cm, height:100%)]  
//Add image over rectangle  
#let arturo = read("Arturo_Nieto-  
Dorantes.webp", encoding: none)  
#image-transparency(arturo, alpha:50%)
```



Parameters

```
image-transparency(  
  imagedata: bytes | path,  
  alpha: ratio,  
  ..args  
) -> content
```

imagedata bytes or path

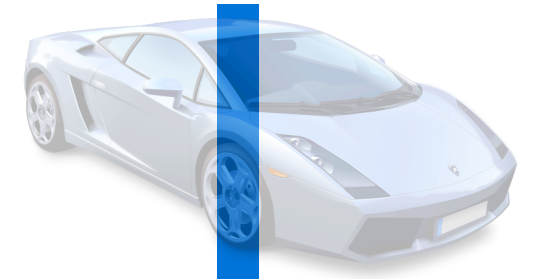
Raw imagedata in any of the supported formats, e.g. provided by the `read()` function or a path to a file

alpha ratio

Remaining amount of visibility

0% = fully transparent, 100% = fully opaque

```
#import "@preview/grayness:0.8.0": *  
//Place rectangle in background to  
demonstrate transparency  
#place(top+center, dx:-3cm)  
[#rect(fill:blue, width:4cm,  
height:100%)]  
#let gallardo = read("gallardo.svg")  
#image-transparency(gallardo, alpha:  
30%, format: "svg")
```



Default: 50%

..args

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

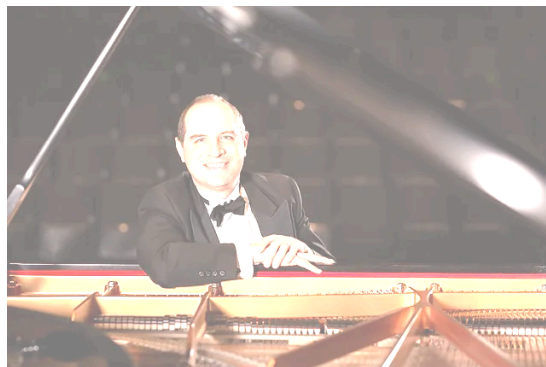
You must pass `format:"svg"` as argument if you use a SVG-image as your input.

image-brighten

Brightens the supplied image

Example:

```
#import "@preview/grayness:0.8.0": *  
#let arturo = read("Arturo_Nieto-  
Dorantes.webp", encoding: none)  
#image-brighten(arturo, amount:50%)
```



Parameters

```
image-brighten(  
  imagedata: bytes | path ,  
  amount: ratio ,  
  ..args: arguments  
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

amount ratio

Amount to brighten by. 0% = no brightening, 100% = completely white

Default: 50%

..args arguments

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

image-darken

Darkenes the supplied image

Example:

```
#import "@preview/grayness:0.8.0": *  
#let arturo = path("Arturo_Nieto-Dorantes.webp")  
#image-darken(arturo, amount:50%)
```



Parameters

```
image-darken(  
  imagedata: bytes | path ,  
  amount: int ,  
  ..args: arguments  
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

amount int

Amount to darken by 0% = no darkening, 100% = completely black

Default: 50%

..args arguments

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-darken(gallardo, amount:50%,  
  format:"svg")
```



image-invert

Inverts the colors of the supplied image

Example:

```
#import "@preview/grayness:0.8.0": *  
#let arturo = read("Arturo_Nieto-  
Dorantes.webp", encoding: none)  
#image-invert(arturo)
```



Parameters

```
image-invert(  
  imagedata: bytes path ,  
  ..args: arguments  
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

..args arguments

Arguments to pass to the typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

```
#import "@preview/grayness:0.8.0": *  
#let gallardo = read("gallardo.svg")  
#image-invert(gallardo, format:"svg")
```

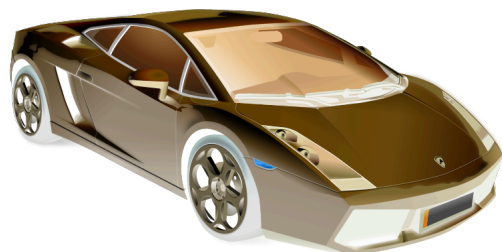
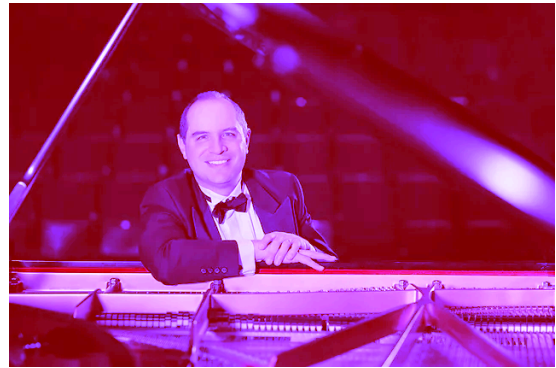


image-matrix

Applies a color-matrix transformation to the supplied image according to the following formula:

$$\begin{aligned}R' &= r1*R + r2*G + r3*B + r4*A + r5 \\G' &= g1*R + g2*G + g3*B + g4*A + g5 \\B' &= b1*R + b2*G + b3*B + b4*A + b5 \\A' &= a1*R + a2*G + a3*B + a4*A + a5\end{aligned}$$

```
#import "@preview/grayness:0.8.0": *
#let arturo = path("Arturo_Nieto-
Dorantes.webp")
#let matrix = (
  (0.5, 0.0, 0.0, 0.0, 0.5),
  (0.0, 1.0, 0.0, 0.0, 0.0),
  (0.0, 5.0, 1.0, 0.0, 0.0),
  (0.0, 0.0, 0.0, 1.0, 0.0),
)
#image-matrix(arturo, matrix)
```



Parameters

```
image-matrix(
  imagedata: bytes | path,
  matrix: array,
  ..args: arguments
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

matrix array

5x4 matrix as array of arrays with floats in the following order:

```
(
  (r1, r2, r3, r4, r5),
  (g1, g2, g3, g4, g5),
  (b1, b2, b3, b4, b5),
  (a1, a2, a3, a4, a5),
)
```

..args arguments

Arguments to pass to the typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

```
#import "@preview/grayness:0.8.0": *
#let gallardo = read("gallardo.svg")
#let matrix = (
  (0.5, 0.0, 0.0, 0.0, 0.5),
  (0.0, 1.0, 0.0, 0.0, 0.0),
  (0.0, -5.0, 1.0, 0.0, 0.0),
  (0.0, 0.0, 0.0, 1.0, 0.0),
)
#image-matrix(gallardo, matrix,
format:"svg")
```

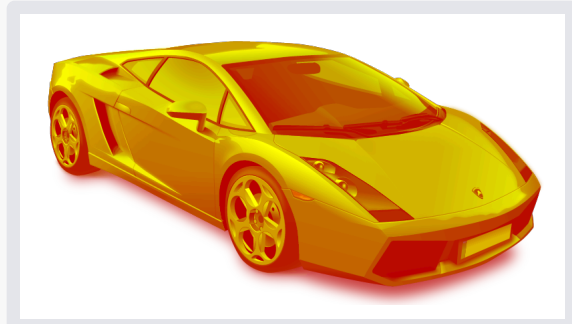


image-huerotate

Hue rotate the supplied image.

Example:

```
#import "@preview/grayness:0.8.0": *
#let arturo = read("Arturo_Nieto-
Dorantes.webp", encoding: none)
#image-huerotate(arturo, amount:100)
```



Parameters

```
image-huerotate(
  imagedata: bytes path,
  amount: int float,
  ..args: arguments
) -> content
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file

amount int or float

Number of degrees to rotate each pixels color by. 0 and 360 do nothing, the rest rotates by the given degree value. Smallest step for raster images is 1°.

Default: 0

..args arguments

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

You must pass `format:"svg"` as argument if you use a SVG-image as your input.

```
#import "@preview/grayness:0.8.0": *
#let gallardo = read("gallardo.svg")
#image-huerotate(gallardo, amount:100,
format:"svg")
```



image-mask

For raster images, this function updates the alpha-channel of the image by applying the masking image to it. The if the mask image is not the same size as the target image, it will be resized automatically.

For SVG, the mask image is applied to the SVG directly without scaling if it has a different aspect ratio. The result image will appear transparent at areas not covered by the mask in this case.

```
#import "@preview/grayness:0.8.0": *
#let arturo = read("Arturo_Nieto-
Dorantes.webp", encoding: none)
#let mask = read("mask.png",
encoding:none)
#image-mask(arturo, mask)
```



Parameters

```
image-mask(
  imagedata: bytes path,
  maskdata: bytes,
  use-alpha-channel: bool,
  ..args: arguments
)
```

imagedata bytes or path

Raw imagedata, e.g. provided by the `read()` function or a path to a file for the source image

maskdata bytes

Raw imagedata, e.g. provided by the `read()` function or a path to a file for the mask image

use-alpha-channel bool

For raster images, defines if the alpha-channel of the mask image is used (default). If set to false, the brightness of the mask image is used instead. Therefore, images without an alpha-channel can also be used as mask.

For SVG images, the brightness and alpha channel are always used both for masking.

```
#import "@preview/grayness:0.8.0": *  
#let arturo = read("Arturo_Nieto-Dorantes.webp", encoding: none)  
#let mask = read("mask.png",  
encoding:none)  
#image-mask(arturo, mask, use-alpha-channel:false)
```



Default: **true**

..args arguments

Arguments to pass to the Typst image function e.g. width, height, alt, fit, ...

image-infos

Gets the images dimensions and its format as string The dimensions are given in pixels for raster images. For SVG, the width, height and viewBox for the toplevel SVG Element are returned if present in the original. They should include units if the file adheres to the SVG 2.0 standard.

```
#import "@preview/grayness:0.8.0": *
#let arturo = path("Arturo_Nieto-
Dorantes.webp")
#let gallardo = path("Arturo_Nieto-
Dorantes.webp")
#let infos_a = image-infos(arturo)
#let infos_g = image-infos(gallardo,
format: "svg")

Width: #infos_a.width, Height:
#infos_a.height, Format: #infos_a.format\
Width: #infos_g.width, Height:
#infos_g.height, Viewbox: #infos_g.viewBox
```

```
Width: 640, Height: 426, Format: WebP
Width: 1901, Height: 1018, Viewbox: 0 0 1901 1018
```

Parameters

```
image-infos(
  imagedata: bytes path,
  ..args
) -> dictionary
```

imagedata bytes or path

Raw imagedata, e.g. provided by the read() function or a path to a file

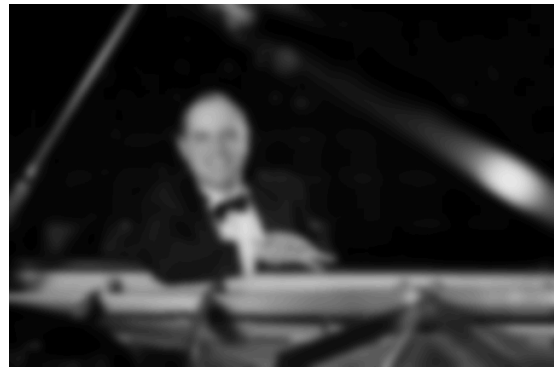
plg

Provides direct access to the webassembly functions, which can be used to chain several operations after one another. The available methods are:

- `grayscale(imagebytes)`
- `svg_grayscale(image_bytes)`
- `convert(image_bytes)`
- `mask(target_image_bytes, mask_image_bytes, use_alpha)`
- `svg_mask(target_image_bytes, mask_image_bytes)`
- `crop(image_bytes, start_x, start_y, width, height)`
- `svg_crop(image_bytes, start_x, start_y, width, height)`
- `blur(image_bytes, sigma)`
- `svg_blur(image_bytes, sigma)`
- `transparency(image_bytes, alpha)`
- `svg_transparency(image_bytes, alpha)`
- `invert(image_bytes)`
- `svg_invert(image_bytes)`
- `brighten(image_bytes, amount)`
- `svg_brighten(image_bytes, amount)`
- `huerotate(image_bytes, amount)`
- `svg_huerotate(image_bytes, amount)`
- `matrix(imagebytes, m00, m01, m02, m03, m04, m10, m11, m12, m13, m14, m20, m21, m22, m23, m24, m30, m31, m32, m33, m34)`
- `svg_matrix(imagebytes, m00, m01, m02, m03, m04, m10, m11, m12, m13, m14, m20, m21, m22, m23, m24, m30, m31, m32, m33, m34)`
- `infos(imagebytes)`
- `svg_infos(imagebytes)`
- `decode(imagebytes)`

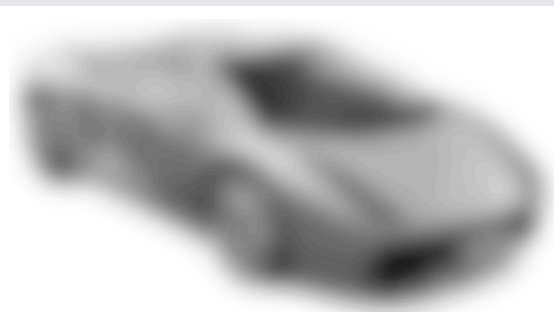
Raster Example:

```
#import "@preview/grayness:0.8.0": plg
#let arturo = read("Arturo_Nieto-
Dorantes.webp", encoding: none)
#let grayscaled-bytes =
plg.grayscale(arturo)
#let blurred-and-grayscaled =
plg.blur(grayscaled-bytes, float(5).to-
bytes(size: 4))
#image(blurred-and-grayscaled)
```



Vector Example:

```
#import "@preview/grayness:0.8.0": plg
#let gallardo = read("gallardo.svg",
encoding: none)
#let grayscaled-bytes =
plg.svg_grayscale(gallardo)
#let blurred-and-grayscaled =
plg.svg_blur(grayscaled-
bytes, float(40).to-bytes(size: 4))
#image(blurred-and-grayscaled)
```



The wasm plugin has its own version number which can be queried using `#str(plg.plugin_version())`. The current plugin version is 0.7.0